Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=5;i++)
    (i-1)[arr] = (i%5);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 3;
}
int main(){
  int arr[6] = {2,2,2,2,2};
  func1(arr+1);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

arr[i-1] = i%5 ;
0 = 0
0 1 2 3 4.

2

5

0 1 2 3 4
0

0 4 8
{1, 2, 3}

50 + 4 = 200

Output
2 0 1 2 3 4
5 0 1 2 3 4

Will the answer change if we write func2 as:
`void func2 (int a[1] , int arr){`
`int temp = arr + 4;`
`a = &temp;`
`}`
Explain your answer

Yes, because originally the actual arr[0] was being changed to 5. Now, pointer to arr[0] is &temp.

Q. 2: Write the output of the following sequence of instructions **long arr2[90]; int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

400 200

The size of long int is 8 bytes and for int is 4 bytes. Also, a1, a2, b1, b2 correspond to the hexadecimal memory addresses. So, a1 - a2 is difference between them.